

AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD

Department of Information Technology

Pre-University Test

Course: B. Tech.

Semester: 4th

Session: 2025-26

Section: IT-1,2,3/CSIT-1,2/CSE& allied branch

Subject: Operating system

Sub. Code: BCS-401

Max Marks: 70

Time: 3 hrs.

OBE Remarks:

Q.No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
CO No.	1	2	2	1	3	4	5	1	2	2	2	3	3	4	4	5	5
Bloom's Level	L1	L2	L3	L1	L1	L2	L1	L3	L4	L2	L5	L4	L3	L4	L4	L2	L5
													L2	L5	L5		L2
Weightage CO3: 16					Weightage CO4: 16					Weightage CO5: 16							

Note: Answer all the sections.

Section-A

A. Attempt all the parts.

(7 X 2 = 14)

1. Define any two functions of operating systems.
2. Explain Context Switching.
3. Illustrate Race Condition by using example.
4. Define time sharing and real time operating system?
5. What are safe state and an unsafe state?
6. Explain the concept of contiguous and non contiguous memory allocation.
7. Define seek time and Latency time.

Section-B

B. Attempt any three. (Q. No. 12 is Compulsory)

(3X 7=21)

8. Explain briefly Layered operating System structure with neat sketch. Also explain protection and security.
9. State Mutual Exclusion, implementation with test and set instruction.
10. For the following processes, draw the Gantt chart to illustrate the execution using,
a) Preemptive priority scheduling b) Non-Preemptive priority scheduling
c) Also calculate average waiting time and average turnaround time. Assume a larger priority number has higher priority.

Process	Arrival Time	Burst Time	Priority
P0	0	4	3
P1	2	5	5
P2	4	3	6
P3	5	2	4
P4	6	6	2

11. Consider the following system with five processes (P1, P2, P3) and three resources (R1, R2, R3)

	Allocated			Maximum			Available		
Process	R1	R2	R3	R1	R2	R3	R1	R2	R3
P1	2	2	3	3	6	8	7	7	10
P2	2	0	3	4	3	3			
P3	1	2	4	3	4	4			

Roll No.																			
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- i) Calculate the content of Need Matrix ii) Apply safety algorithm and check the current system is in safe state or not iii) discuss prevention and avoidance techniques
12. Explain the analogy of Dining philosopher problem with respect to computer system. Also provide its probable solution.

Section-C

(5 X 7 = 35)

C. Attempt all the parts.

13. Attempt any one.

- a) Illustrate Peterson and Dekker Problem with solutions?
- b) Define Semaphores in detail and also illustrate any problem that will solve by Semaphores with their operations.

14. Attempt any one.

- a) Difference between logical address and physical address space? On a simple paging system with 232 frames of physical memory, 212 pages of virtual address space and page size of 64 bytes, how many bits are in a. Virtual Address b. Physical address c. Page number d. Frame number e. Offset
- b) Difference between paging and segmentation? Given LA=24b, PA=16b, PS= 1kb so find out how many bits in Page number, how many pages in secondary memory and main memory? Also find the bits of displacement of LA and PA?

15. Attempt any one.

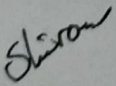
- a) Consider the following page reference string: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6 How many page faults occur for the following page replacement algorithms assuming three frames? (i) FIFO (ii) LRU (iii) Optimal
- b) Given memory fixed size partitions of 100K, 500K, 200K, 300K, and 600K (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 212K, 417K, 112K and 426K (in order)? Which algorithm makes the most efficient use of memory?

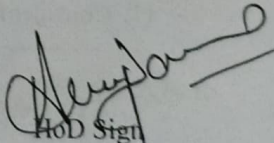
16. Attempt any one.

- a) Describe the File System Implementation and Directory System Implementation in detail
- b) Explain RAID and its various levels with their advantages and disadvantages.

17. Attempt any one.

- a) A hard disk is having 2000 cylinders, numbered 0 to 1999, the drive is currently serving the request at cylinder 143 and the previous request was at cylinder 125. The status of the queue is as follows 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 What is the total distance in cylinders that the disk arm moves to satisfy the entire pending request for each of the following disk-scheduling algorithms?
- i. SSTF ii. SCAN iii. CSCAN iv. LOOK
- b) Write short notes on:
- a. File system protection and security b. Linked file allocation methods c. Direct Memory Access (DMA).


Faculty Sign


HoD Sign